

Mixed-model association under population structure: numerical foundations, calibration diagnostics and research directions for mixmogam

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Abstract

Mixed-model association combines a model of genetic background with a numerical approximation and a reference distribution for testing. These three parts require separate validation. We describe the revised mixmogam package and examine how population structure exposes their differences. The package provides dense and streaming Gaussian mixed-model analyses, leave-one-chromosome-out (LOCO) scans, and experimental two-step approximations. We derive the dependence of the association denominator on a variant’s orientation in the kinship spectrum and show why calibration averaged over variants can conceal opposing errors. In a new conditional experiment with known covariance, 256 samples, and 970 held-out audit variants, scalar calibration gives an overall rejection probability of 1.00% at a nominal 1% threshold under moderate frequency dispersion, but 1.19% and 0.73% in the lowest and highest structure-loading quintiles. A rank-four spectral correction reduces that spread to 1.010% and 1.005%. Analytic probabilities agree with simulations across 2,000 independent phenotype replicates. These results establish a mechanism under a controlled Gaussian null; they do not validate estimated-covariance scans or reference GWAS programs. We reassess historical simulations, separate environmental mean misspecification from denominator error, and withdraw an external comparison compromised by genotype encoding. A reproducible research agenda prioritizes matched-input benchmarks, independent calibration audits, error-controlled spectral approximations, and validation under realistic ancestry, relatedness, and genetic architectures.

1 Introduction

A mixed-model association test asks whether a candidate variant explains phenotypic variation beyond specified covariates and a model of genetic background. Its appeal is that the same relationship matrix can describe close relatives, cryptic relatedness, and broad population structure. Early efficient implementations made this approach practical in model organisms and human association studies (Kang et al., 2008, 2010; Lippert et al., 2011; Zhou and Stephens, 2012). The resulting test is nevertheless conditional on assumptions about the phenotype mean, the background covariance, and the way nuisance parameters are fitted. A precise matrix calculation cannot repair a misspecified mean, and a well-fitted prediction model does not by itself supply a valid association test.

The computational problem has changed with cohort size. Dense methods factorize an $n \times n$ covariance matrix and reuse the factorization across variants. Large-cohort methods instead use iterative fitting, whole-genome prediction, and approximate association statistics. BOLT-LMM

combines mixed-model association with a mixture prior on marker effects (Loh et al., 2015); REGENIE constructs whole-genome predictions through successive ridge regressions (Mbatchou et al., 2021); and newer approaches include LDAK-KVIK (Hof and Speed, 2025) and the variational method Quickdraws (Loya et al., 2025). These programs have distinct fitting and testing procedures. A claim about one denominator approximation should therefore be evaluated at the level of that approximation, rather than assigned to all methods with a two-step workflow.

Two statistical choices cut across these computational designs. First, a candidate variant and its correlated neighbours can contribute to the random-effect predictor against which they are tested. This proximal contamination can reduce power; excluding the candidate chromosome or region changes that overlap (Listgarten et al., 2012; Yang et al., 2014). Second, ancestry can influence both allele frequencies and environmental exposures. Modelling genetic covariance and adjusting the phenotype mean with principal components (PCs) address different parts of this problem (Price et al., 2006; Vilhjálmsón and Nordborg, 2013). Neither chromosome exclusion nor covariance modelling makes environmental confounding disappear as a matter of algebra.

This report focuses on a third, more specific issue: the denominator of an association statistic depends on how the tested variant aligns with the covariance eigenvectors. Replacing that denominator by a scalar multiple of the variant norm can work well when this alignment has little effect. Under structure, a satisfactory average can instead hide inflation in one class of variants and deflation in another. The relevant object is the conditional rejection probability for a variant, or a prespecified class of variants, under a stated null model. A single genome-wide median is an incomplete diagnostic of that object.

mixmogam offers a useful setting for studying these distinctions because dense and streaming calculations share genotype handling and several model components. Its earlier work included multi-locus mixed-model association (Segura et al., 2012); the current engine also supplies LOCO and experimental two-step analyses. The present contribution is a methods and software investigation, not a competitive benchmark establishing a new best-performing GWAS program. We provide the mathematical contracts needed to interpret its outputs, a new known-covariance experiment with a disjoint set of audit variants, and a critical reading of archived simulations. We then formulate research questions that can distinguish a useful general method from an approximation that works only in a favourable example.

The evidence has three levels. Current software checks assess specified numerical and data-handling contracts. The new experiment isolates calibration geometry under a deliberately simple null. Historical simulations motivate broader questions but predate the scientific correctness review. Table 1 records these boundaries; they apply throughout the report.

2 Statistical and computational methods

2.1 Data, estimand, and relationship matrix

Let $\mathbf{y} \in \mathbb{R}^n$ be a quantitative phenotype, $\mathbf{X}$ a full-rank $n \times p$ covariate matrix including an intercept, and $\mathbf{z}_j$ a centered candidate genotype. The coefficient of interest, α_j , is the fixed effect in a specified marginal mixed model after integrating out the random genetic background. It is not automatically a direct causal effect: linkage disequilibrium (LD), omitted exposures, and the choice of background model affect its interpretation. The null is $H_{0j} : \alpha_j = 0$ under that model. A variant without a designated large QTL effect need not satisfy this null if a simulated polygenic background still assigns it an effect.

The relationship matrix summarizes the covariance induced by a polygenic marker model (Yang et al., 2010). For the unweighted matrix, mixmogam centers each marker by its mean among called

Table 1: Evidence used in this revision and the inference it supports. Historical archives are preserved, not silently reclassified as results from the corrected package.

Evidence	Established here	Remaining boundary
Current review, 2.0.0.dev1	Regression and independent numerical checks; 117 local tests passed; installed-artifact checks	Test coverage does not establish population calibration.
New denominator experiment	Analytic conditional rejection probabilities, checked by independent phenotype replicates	Known Gaussian covariance, small fixed genotype panels, exact leading eigenvectors.
Archived simulations	Descriptive patterns under the recorded historical implementation	New input handling and fitting changes need fresh end-to-end runs.
Archived external comparison	A provenance failure was identified: incorrect PLINK export encoding	External-program calibration, agreement, and superiority claims are withdrawn pending a matched-input rerun.

samples and divides by its empirical standard deviation among those samples. Missing calls become zero after this transformation, corresponding to mean imputation. Write the resulting background matrix as $\mathbf{Z}$, with M retained polymorphic markers. Its raw relationship matrix and a weighted extension are

$$\mathbf{K}_0 = \frac{1}{M} \mathbf{Z} \mathbf{Z}^\top, \quad \mathbf{K}_w = \frac{\sum_{j=1}^M w_j \mathbf{z}_j \mathbf{z}_j^\top}{\sum_{j=1}^M w_j}, \quad w_j \geq 0, \quad \sum_j w_j > 0. \quad (1)$$

Empirical-standard-deviation scaling differs from using the Hardy–Weinberg variance $2f_j(1 - f_j)$; these definitions should not be interchanged in a reference comparison. The optional kinship rescaling subtracts the mean off-diagonal entry from every entry and divides by the resulting mean diagonal. An intercept absorbs the constant-matrix shift in the residual subspace. The raw, unrescaled matrix is used in the new experiment in Section 3.1.

The current input contract is diploid hard calls 0, 1, 2, with -1 for missing values. It is not a continuous-dosage interface. Minor allele count is measured in allele copies. Sample identifiers must align uniquely; silently intersecting phenotype and genotype rows in different orders changes the scientific problem. PLINK effect-allele orientation, missingness, and identifier preservation are consequently part of the statistical contract, not merely file-format details.

2.2 Gaussian model, variance fitting, and chromosome exclusion

For a candidate in chromosome group g , the working model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}_j\alpha_j + \mathbf{u} + \mathbf{e}, \quad \mathbf{u} \sim N(0, \sigma_g^2 \mathbf{K}_{-g}), \quad \mathbf{e} \sim N(0, \sigma_e^2 \mathbf{I}), \quad \mathbf{u} \perp \mathbf{e}. \quad (2)$$

Here $\mathbf{K}_{-g}$ is constructed without the markers in group g . The subscript on σ_g^2 denotes genetic variance, not the group index. With $\delta = \sigma_e^2 / \sigma_g^2$, set $\mathbf{H} = \mathbf{K}_{-g} + \delta \mathbf{I}$ and $\mathbf{V} = \sigma_g^2 \mathbf{H}$. This parameterization assumes a positive genetic scale; the zero-genetic-variance limit reduces to a linear model.

The covariance shape is fitted under the null $\alpha_j = 0$. For fixed δ , define the residual precision matrix

$$\mathbf{P}_H = \mathbf{H}^{-1} - \mathbf{H}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{H}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{H}^{-1}. \quad (3)$$

Up to constants independent of δ , the profiled restricted log-likelihood is

$$\ell_R(\delta) = -\frac{1}{2} \left\{ \log |\mathbf{H}| + \log |\mathbf{X}^\top \mathbf{H}^{-1} \mathbf{X}| + (n-p) \log \left(\frac{\mathbf{y}^\top \mathbf{P}_H \mathbf{y}}{n-p} \right) \right\}. \quad (4)$$

Thus $\hat{\sigma}_g^2 = \mathbf{y}^\top \mathbf{P}_H \mathbf{y} / (n-p)$ for REML at the fitted δ . Maximum likelihood instead uses n and omits the covariate determinant term. Confusing these objectives changes both the fit and its interpretation. A phenotype entirely explained by $\mathbf{X}$ has zero residual variance and is rejected rather than assigned numerical noise as a variance estimate.

LOCO matrices can be formed by subtracting group-specific cross-products from the genome-wide sum. The dense implementation refits variance components for each group; the streaming infinitesimal path uses one global variance fit and group-specific solves. These are distinct nuisance-fitting procedures. When chromosome labels are numerous, the implementation can merge them into a limited number of groups, so the actual grouping must accompany the results. Exclusion reduces direct overlap between the tested region and its background predictor; it does not imply statistical independence across groups or correct an omitted phenotype mean.

2.3 What an exact scan makes exact

For a fixed covariance shape $\mathbf{H}$, let $U_j = \mathbf{z}_j^\top \mathbf{P}_H \mathbf{y}$, $q_j = \mathbf{z}_j^\top \mathbf{P}_H \mathbf{z}_j$, and $S_0 = \mathbf{y}^\top \mathbf{P}_H \mathbf{y}$. Generalized least squares gives

$$\hat{\alpha}_j = \frac{U_j}{q_j}, \quad F_j = \frac{U_j^2 / q_j}{(S_0 - U_j^2 / q_j) / (n-p-1)}. \quad (5)$$

Under a Gaussian null with this *fixed, correct* covariance shape, $q_j > 0$, and positive residual degrees of freedom, $F_j \sim F_{1, n-p-1}$. In EMMAX-style scans, however, δ is first estimated from the phenotype and then held fixed while variants are tested (Kang et al., 2010). The same algebra can be computed accurately, but the finite-sample reference law is then a plug-in approximation. Throughout this report, “exact scan” refers to the dense calculation with its fitted covariance, not a guarantee of exact type-I error.

The corresponding known-covariance score test is particularly useful for analysis. Replace $\mathbf{H}$ by the fully specified $\mathbf{V}$ in Equation (3) to obtain $\mathbf{P}_V$. Since $\mathbf{P}_V \mathbf{V} \mathbf{P}_V = \mathbf{P}_V$, the null statistic satisfies

$$T_j = \frac{(\mathbf{z}_j^\top \mathbf{P}_V \mathbf{y})^2}{\mathbf{z}_j^\top \mathbf{P}_V \mathbf{z}_j} \sim \chi_1^2. \quad (6)$$

This conditional result is the reference for the new experiment. It avoids conflating an approximation to the denominator with uncertainty in estimated variance components. Variants collinear with the covariates have zero residual norm and are not estimable.

2.4 Scalar calibration and its spectral meaning

Work in orthonormal coordinates spanning the complement of $\mathbf{X}$. Let $R^\top R = \mathbf{I}_d$, $R^\top \mathbf{X} = 0$, and $d = n-p$; use $\tilde{\mathbf{y}} = R^\top \mathbf{y}$, $\tilde{\mathbf{z}}_j = R^\top \mathbf{z}_j$, and $\tilde{\mathbf{K}} = R^\top \mathbf{K} R$. For the rest of this subsection the tildes are suppressed. The infinitesimal solve is $\mathbf{w} = (\mathbf{K} + \delta \mathbf{I})^{-1} \mathbf{y}$. A retrospective statistic uses $D_j^{\text{ret}} = (\mathbf{z}_j^\top \mathbf{z}_j / d) \|\mathbf{w}\|^2$ in the denominator of $(\mathbf{z}_j^\top \mathbf{w})^2$. The matching prospective denominator is $D_j^{\text{pro}} = \sigma_g^2 q_j$, where $q_j = \mathbf{z}_j^\top (\mathbf{K} + \delta \mathbf{I})^{-1} \mathbf{z}_j$. A calibration factor averages $D_j^{\text{pro}} / D_j^{\text{ret}}$ over selected markers. A common factor can reproduce the prospective statistic for all markers only when the relevant ratios are constant.

Write $\mathbf{K} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $a_{\ell j} = \mathbf{U}_\ell^\top \mathbf{z}_j$. The missing geometry is explicit:

$$q_j = \sum_{\ell=1}^d \frac{a_{\ell j}^2}{\lambda_\ell + \delta}, \quad \|\mathbf{z}_j\|^2 = \sum_{\ell=1}^d a_{\ell j}^2. \quad (7)$$

A direction assigned high background variance receives low precision. Two variants with equal norms can therefore have different q_j . A denominator calibrated to the average may be too small for variants concentrated in low-variance directions and too large for variants concentrated in high-variance directions.

For known covariance and a deterministic approximate denominator $\hat{q}_j$, put $r_j = q_j/\hat{q}_j$. The consequence is exact:

$$\tilde{T}_j = r_j T_j, \quad \Pr(\tilde{T}_j > c_\alpha \mid \mathbf{Z}) = \bar{F}_{\chi_1^2} \left(\frac{c_\alpha}{r_j} \right), \quad c_\alpha = F_{\chi_1^2}^{-1}(1 - \alpha). \quad (8)$$

An error that appears modest near the center can matter much more in the tail. Averaging Equation (8) over a variant group gives its expected rejection fraction, without assuming variants are independent. It is a marginal rejection probability, not the family-wise probability of at least one discovery.

2.5 A low-rank correction and its current limits

Retaining the leading k eigenpairs and replacing the rest by their mean $\bar{\lambda}_{>k}$ gives the approximation

$$q_j^{(k)} = \sum_{\ell \leq k} \frac{a_{\ell j}^2}{\lambda_\ell + \delta} + \frac{\|\mathbf{z}_j\|^2 - \sum_{\ell \leq k} a_{\ell j}^2}{\bar{\lambda}_{>k} + \delta}, \quad \hat{q}_j = c_k q_j^{(k)}. \quad (9)$$

The scalar c_k corrects average remaining error; it cannot remove arbitrary variation in the spectral tail. For a LOCO analysis, each approximation should target the covariance used by that group's test. Using a full-genome basis to approximate a different LOCO covariance requires its own error assessment.

The production implementation computes group-specific bases with randomized subspace iteration and currently tries widths 64, 128, 256, and 512. It stops when the coefficient of variation of calibration ratios is at most 3%, subject to the available dimension. The calibration markers also determine the scale, so this stopping criterion is an internal diagnostic. It is not an independent validation and does not bound the error at genome-wide thresholds. The production infinitesimal calibration also selects candidate markers using a phenotype-dependent statistic; this differs from the phenotype-independent selection in our controlled experiment.

For mixture-prior and elastic-net residuals, an infinitesimal spectral factor is transferred before further calibration. Once the residual is a nonlinear, data-adaptive function of $\mathbf{y}$, the proof of Equation (6) no longer supplies its null law. This transfer is a heuristic. The names “BOLT-style” and “KVIK-style” below denote mixmogam implementations inspired by the corresponding methods; they do not establish equivalence to the reference programs.

2.6 Environmental mean misspecification

Suppose the actual conditional mean includes an omitted exposure vector $\mathbf{d}$ with effect γ : $\mathbb{E}(\mathbf{y} \mid \mathbf{Z}, \mathbf{d}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{d}\gamma$. Even if the covariance $\mathbf{V}$ is known correctly, the numerator in Equation (6) then

has mean $\mathbf{z}_j^\top \mathbf{P}_V \mathbf{d} \gamma$. Its distribution becomes

$$T_j \sim \chi_1^2(\eta_j), \quad \eta_j = \frac{(\mathbf{z}_j^\top \mathbf{P}_V \mathbf{d} \gamma)^2}{\mathbf{z}_j^\top \mathbf{P}_V \mathbf{z}_j}. \quad (10)$$

A kinship matrix can assign large variance to an ancestry direction, but it does not automatically place an environmental mean shift in $\mathbf{X}$. Adding covariates that span $\mathbf{d}$ makes the numerator mean zero under this model. Estimated PCs may only approximate that span, and neither the required number of PCs nor a power-preserving adjustment follows from this identity alone. Reviews of structure and mixed models place this distinction in a broader setting (Price et al., 2010; Sul et al., 2018; Yang et al., 2014).

2.7 Numerical implementation and ancillary inference

Dense scans whiten genotype blocks and use matrix products to evaluate Equation (5). Dense factorization costs $O(n^3)$ time and $O(n^2)$ storage, with a dense transformation cost of order $n^2 m$ for m tested variants. Streaming methods replace stored kinships by products with standardized genotype blocks. A product with r right-hand sides costs order nMr arithmetic; its memory consists of the right-hand sides, working blocks, and configured caches. Avoiding an $n \times n$ kinship does not mean constant memory: a genotype cache can still approach the size of the standardized data.

Conjugate gradients supplies covariance solves, and stochastic Lanczos quadrature estimates likelihood terms for large samples. Both introduce numerical approximations beyond the statistical model. Association solves now reject nonconvergence. Variational fits expose incomplete convergence, and fit caches respect their options and model ownership. The review also removed an unnecessary LOCO-group axis from temporary products; Section 4.4 reports the measured allocation change with its scope.

Several adjacent routines need equally explicit contracts. The genetic BLUP is $\hat{\mathbf{u}} = \mathbf{K} \mathbf{H}^{-1} (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}})$; the genetic scale cancels and must not be multiplied in a second time. Default permutation inference operates on orthogonal residual coordinates rather than permuting the correlated entries of a projected residual vector, following the underlying distinction developed by Abney (2015). Exact exchangeability requires the stipulated Gaussian covariance; substituting fitted variance components still requires calibration checks. Finally, the normal-approximation Bayes factor requires effect estimates, standard errors, and an explicit effect-variance prior (Wakefield, 2009). It is not a replacement for a joint fine-mapping model. Appendix A gives these identities and their interpretation.

3 Evaluation design and provenance

3.1 New conditional experiment with known covariance

The new experiment isolates Equation (8); it does not call an end-to-end GWAS scan. We generate 256 diploid samples in four equal populations, with 2,000 background and 1,000 candidate markers. For each marker, the ancestral frequency f_j is uniform on $[0.1, 0.9]$. Population-specific frequencies are independent draws from $\text{Beta}(f_j(1/\theta - 1), (1 - f_j)(1/\theta - 1))$ for $\theta > 0$; at $\theta = 0$ they all equal f_j . Hard calls are binomial with two trials. The values $\theta \in \{0, 0.05, 0.20\}$ control frequency dispersion. We do not report them as measured F_{ST} , whose estimation and interpretation require a separate definition (Bhatia et al., 2013).

Markers are independent conditional on population frequencies and sample membership. Population structure nevertheless correlates their sample-level patterns. Monomorphic markers are

removed: one background marker is dropped in the $\theta = 0.20$ panel; all 1,000 candidates remain in every panel. The package’s raw relationship matrix is checked against a direct standardized cross-product. We remove an intercept in orthonormal residual coordinates, set $\delta = 1$ and genetic scale one, and use a full eigendecomposition to obtain the known covariance. Direct linear solves and spectral denominators agree to numerical precision.

Thirty randomly selected candidates calibrate each approximation; the other 970 are held out for audit. Selection is independent of phenotype. The scalar denominator is the variant norm times the mean $q_j/\|\mathbf{z}_j\|^2$ among calibration markers. For the spectral method, we try $k \in \{4, 8, 16, 32, 64, 128\}$ and select the first width whose calibration-ratio coefficient of variation is at most 3%. All three panels select $k = 4$. These exact leading eigenvectors provide a favourable test of spectral geometry, not a validation of the production randomized eigensolver or its width-selection procedure.

Audit variants are divided into quintiles by the share of their squared norm explained by the top three background eigenvectors. For each method and quintile we calculate the analytic mean rejection probability at $\alpha = 0.01$ and 5×10^{-8} . We also simulate 2,000 independent Gaussian phenotype replicates per fixed panel and compare the empirical rejection fraction at 1% with its analytic value. Methods use the same phenotypes. Monte Carlo standard errors are calculated from the replicate-level fractions, retaining correlation among variants within each replicate. We do not estimate the extreme-tail probability by counting a handful of simulated exceedances.

The three genotype panels use seeds 20261003–20261005. Consequently, differences across θ also include differences among three realized panels; there are no genotype-replicate confidence intervals. The analytic result is exact conditional on each realized panel and its fixed approximation. It is not an estimate of an average over all possible populations.

3.2 Historical structure and environmental experiments

The archived structure experiment used 1,300 samples, 20,000 markers, and five chromosomes, with both unstructured genotypes and a four-population generator labelled $F_{ST} = 0.3$ in the archive. We retain that label as a generator setting, not a newly estimated differentiation measure. Six phenotype replicates shared each genotype panel. Genetic signal was generated outside the last chromosome from a polygenic component of variance share 0.4 and ten large QTLs of variance share 0.1. The last chromosome has no generating marker effects, making it useful for a null-chromosome diagnostic. Sparse QTLs, realized cross-chromosome correlations, and variance estimation mean that this is not the known-Gaussian-covariance experiment above. Loading quintiles use the leading ten kinship eigenvectors.

The archived coalescent experiments used mutation and recombination rates 10^{-8} per base pair per generation, effective population size 10^4 , and minor allele frequency above 1% (Kelleher et al., 2016; Baumdicker et al., 2022). Scenario S2 contains two demes separated 400 generations earlier, 400 diploids per deme, and 100 independently simulated blocks of 200 markers. Its environmental exposure is the standardized deme indicator. It is generated from population membership rather than constructed as a PC of the tested genotypes; it is nevertheless strongly correlated with genotype. The raw trait and standardized exposure are each multiplied by $\sqrt{1/2}$, added, and then standardized together. The original genetic variance target of 0.25 becomes 0.125 before final standardization; the environmental variance target is 0.5. Realized component variances and cross-covariances can change their shares of the final phenotype variance. This is a strong quantitative-trait stress test, not a binary liability model or a typical confounding level.

We report S2 statistics in blocks without a designated large QTL. Those blocks still contain infinitesimal effects. We therefore call them *QTL-free blocks*, not strict-null blocks, and do not interpret their detection count as a formal false-positive rate. The same qualification applies to S1

and S3, which have $n = 800$ and $4,000$, $m = 20,000$ and $50,000$, and 20 and 30 large QTLs, respectively. Their target genetic variance share is 0.5, divided equally between a polygenic background and large QTLs. Each scenario has only three independent simulation seeds.

3.3 Summaries, exclusions, and reproducibility

Genomic inflation is summarized as the median chi-square statistic divided by the median of χ_1^2 , approximately 0.455. This statistic mixes model error, true signal, and sampling variation; it is not an estimand of confounding alone. Loading-stratified rejection fractions are more informative about the denominator hypothesis. For the six historical phenotype replicates, Figure 2 uses Student- t intervals for the mean with five degrees of freedom. These intervals concern variation among phenotypes conditional on a fixed genotype panel; six replicates offer limited precision. S2 displays all three seeds as well as their means.

The archived comparison with the external LDAK program used a faulty PLINK encoding path. A matched-input claim cannot survive that failure. We exclude its empirical results, concordance coefficients, and real-data calibration curves from this revision. The underlying RegMap resource (Horton et al., 2012) remains suitable for a future controlled benchmark, but neither its archived export nor a redraw of its old results validates the corrected package.

New numerical results record the generating script, seeds, software versions, requested BLAS thread limits, genotype hashes, and package baseline commit. Figure generation records hashes of its actual inputs and outputs. Historical archives retain their original source snapshots; we do not attach the current version number to computations performed by an earlier implementation.

4 Results

4.1 A nearly correct average can conceal conditional error

Table 2 and Figure 1 show the new conditional experiment. With no frequency differentiation, the scalar approximation is close to the known-covariance test: its analytic rejection probability is 0.99% overall and 1.00% versus 0.97% in the extreme loading quintiles. At $\theta = 0.05$, the overall value remains 1.00% after rounding, but the extreme quintiles separate to 1.19% and 0.73%. The average has become reassuring for the wrong reason: deviations in opposite directions largely cancel.

At $\theta = 0.20$, the scalar approximation rejects 1.25% overall, with 2.21% in the lowest loading quintile and 0.31% in the highest. The corresponding analytic rejection probabilities at a nominal 5×10^{-8} are 27.4 and 0.025 times nominal. These tail ratios follow from the fully specified Gaussian model and Equation (8); they are not empirical estimates of real-cohort false-positive rates. They demonstrate why a denominator should be assessed at the intended testing threshold.

The rank-four correction substantially reduces the loading gradient in these panels. At $\theta = 0.05$, its extreme-quintile probabilities are 1.010% and 1.005%; at $\theta = 0.20$, they are 0.998% and 0.992%. Its overall analytic rejection ratios at 5×10^{-8} range from 0.97 to 1.06 across the three panels. Held-out root-mean-square relative denominator-ratio errors are 1.36–1.53%. This is evidence that the leading structure directions explain most of the approximation error in these particular designs. It neither proves that four directions suffice elsewhere nor tests the accuracy of an estimated leading subspace.

The empirical rejection fractions agree with their analytic targets within the script’s stated numerical and Monte Carlo tolerance. For the exact denominator, the overall empirical fractions range from 1.006% to 1.007%, with standard errors approximately 0.010 percentage points. There

Table 2: New known-covariance experiment. Rejection probabilities at a nominal 1% threshold, expressed as percentages. “Simulated” is the overall mean $\pm$ one Monte Carlo standard error across 2,000 independent phenotype replicates. q1 and q5 are analytic probabilities in the extreme loading quintiles among 970 held-out audit variants.

θ	Denominator	All audit variants		Analytic	
		Analytic	Simulated	q1	q5
0	Exact	1.00	1.01 ± 0.01	1.00	1.00
0	Scalar	0.99	1.00 ± 0.01	1.00	0.97
0	Spectral, rank 4	0.99	0.99 ± 0.01	0.99	0.99
0.05	Exact	1.00	1.01 ± 0.01	1.00	1.00
0.05	Scalar	1.00	1.01 ± 0.01	1.19	0.73
0.05	Spectral, rank 4	1.01	1.02 ± 0.01	1.01	1.01
0.2	Exact	1.00	1.01 ± 0.01	1.00	1.00
0.2	Scalar	1.25	1.26 ± 0.01	2.21	0.31
0.2	Spectral, rank 4	1.00	1.00 ± 0.01	1.00	0.99

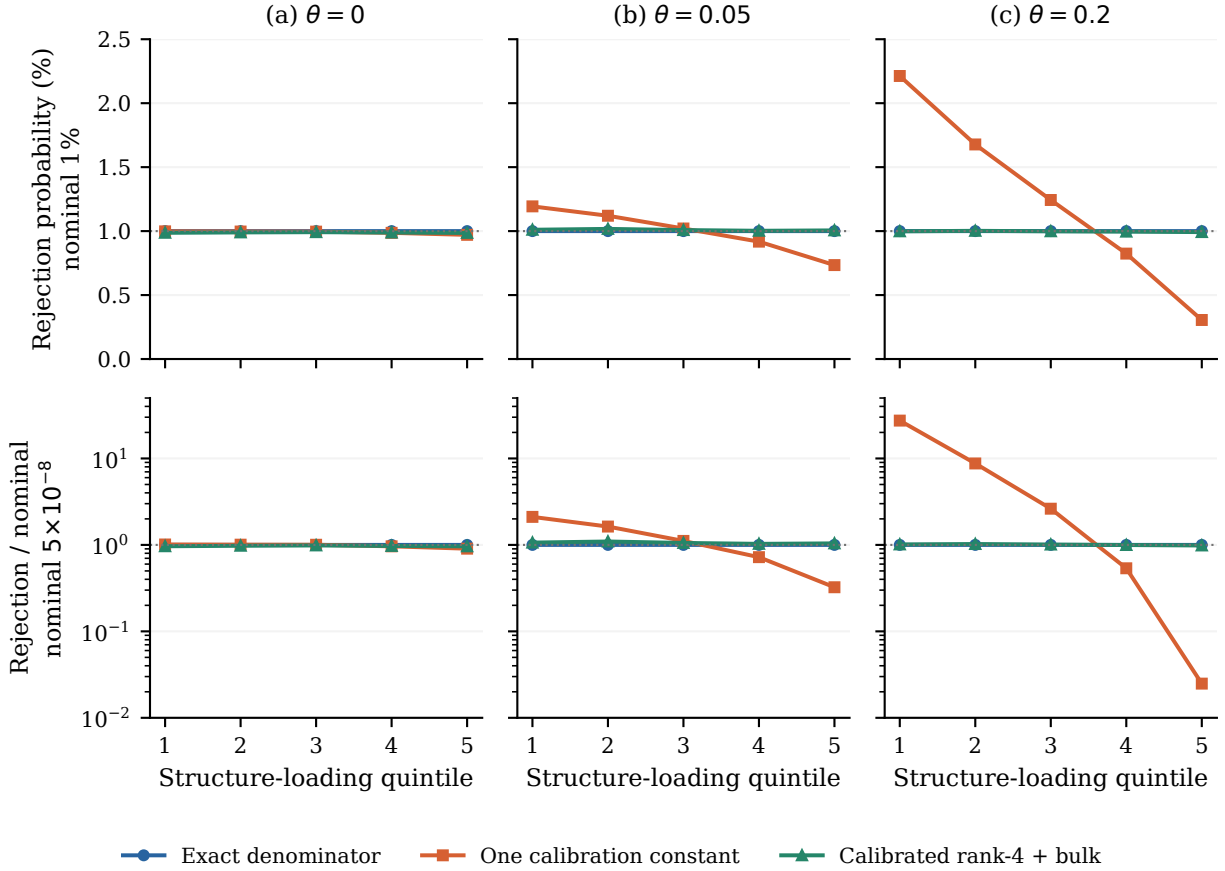


Figure 1: Conditional calibration under known covariance. The top row shows analytic rejection probabilities at a nominal 1%; the bottom row shows the ratio of analytic to nominal rejection probability at 5×10^{-8} on a log scale. Each column is one fixed genotype panel. Candidate variants are held out from approximation calibration and ordered by loading on the leading three background eigenvectors. The spectral method uses four exact eigenvectors and a flat bulk. Columns share vertical scales. Panels describe denominator geometry, not fitted-covariance or external-program performance.

is no fitting uncertainty in this comparison: a score with an exact denominator must have the stipulated marginal law. The simulation checks the implementation of that construction and quantifies the sampling variation of the reported summaries.

4.2 Historical scans show a related pattern, with weaker identification

The archived strong-structure scan exhibits the same qualitative gradient (Table 3, Figure 2). For the scalar infinitesimal implementation, overall λ_{GC} is 0.97, but the lowest and highest loading quintiles have $\lambda_{GC} = 1.34$ and 0.64, and rejection fractions of 2.45% and 0.11% at 1%. The historical spectral correction gives 1.14% and 1.06% in those quintiles, close to the dense LOCO results of 1.12% and 1.11%. The KVIK-style implementation retains a pronounced gradient.

Table 3: Historical simulated strong-structure panel, six phenotype replicates. q1 and q5 denote loading quintiles. Rejection fractions are for the chromosome without generating effects, under the qualifications in Section 3.2. These are archived implementation results, not a rerun of version 2.0.0.dev1 or reference programs.

Method	$\lambda_{GC, \text{ all}}$	λ_{GC}		Rejection (%)	
		q1	q5	q1	q5
Exact LOCO	1.00	1.05	1.02	1.12	1.11
Infinitesimal, constant	0.97	1.34	0.64	2.45	0.11
Infinitesimal, spectral	1.00	1.05	1.01	1.14	1.06
KVIK-style	1.02	1.37	0.73	2.92	0.11

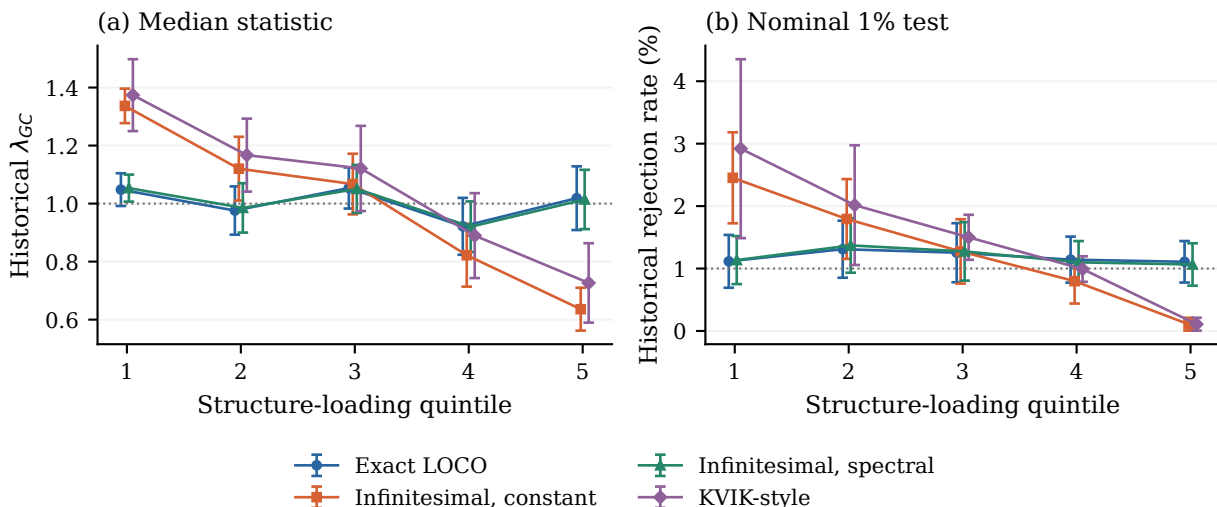


Figure 2: Historical strong-structure diagnostics. Points are means over six phenotype replicates on one fixed genotype panel; bars are 95% Student- t intervals for those means. The median statistic and the nominal 1% rejection fraction show why reporting only a genome-wide summary can obscure differences among variant groups. No external program results are included.

These results are supportive diagnostics, not a second proof of the mechanism. Variance fitting, sparse generating effects, calibration-marker selection, and the earlier implementation all contribute to the final statistics. The new experiment is stronger for identifying denominator error precisely

because it holds those features fixed or removes them. Conversely, its simplicity limits what it says about the complete scan. Both evidence types are useful when their roles are kept distinct.

4.3 Covariance adjustment and environmental adjustment are different

In the historical S2 stress test, the leading PC predicts deme membership with squared correlation approximately 0.97. Without kinship adjustment, QTL-free blocks have $\lambda_{GC} = 11.90$; the mean statistic among variants with the highest environmental loading is 63.24. Dense LOCO reduces these summaries to 1.24 and 2.27, but substantial residual structure remains (Table 4, Figure 3). Including PC1 reduces them further to 1.09 and 1.13. The corresponding all-variant λ_{GC} falls from 1.27 to 1.11. The residual is reduced, not demonstrably eliminated.

Table 4: Historical S2 environmental stress test, means over three seeds. Loading refers to squared correlation with the deme exposure. Detection counts are significant QTL-free blocks, not false positives under a strict null; these blocks also contain polygenic background effects. Power is the fraction of designated QTL-containing loci detected under the archived Bonferroni rule.

Method	λ_{GC} , QTL-free	Mean χ^2		QTL-free detections	Locus power
		Bottom half	Top 1%		
Linear model	11.90	2.20	63.24	90.33	1.00
LMM, no LOCO	0.96	0.87	1.77	0.00	0.13
Exact LOCO	1.24	1.13	2.27	0.67	0.20
Exact LOCO + PC1	1.09	1.10	1.13	0.67	0.20

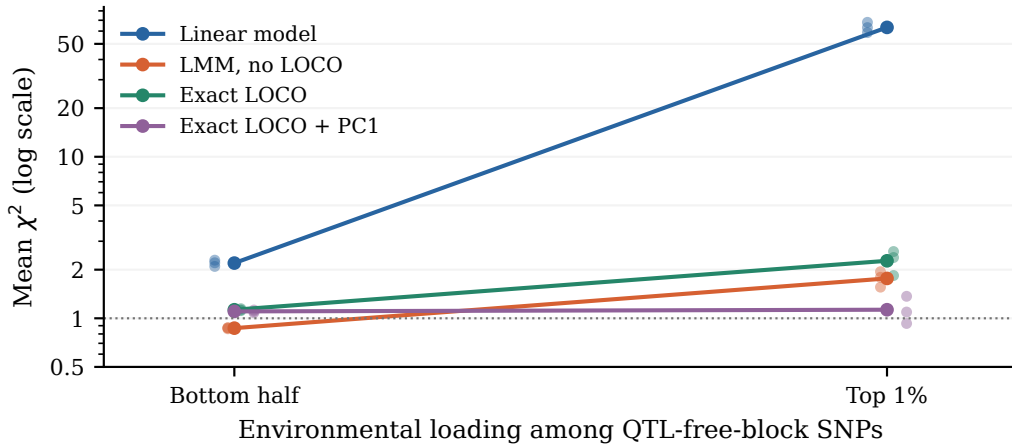


Figure 3: Historical S2 statistics by environmental loading in QTL-free blocks. Large points and lines show means; small points show the three simulation seeds. A logarithmic vertical axis accommodates the large unadjusted mean. PC adjustment reduces the observed gradient, but the polygenic background and small replicate count prevent a strict-null calibration claim.

The non-LOCO scan has all-variant $\lambda_{GC} = 0.98$ and QTL-free $\lambda_{GC} = 0.96$, yet its upper-loading mean remains 1.77 and its locus power is lower than LOCO’s (0.13 versus 0.20). Apparent aggregate calibration is therefore compatible with both residual environmental structure and signal attenuation. These data do not identify their separate causal contributions; a factorial simulation holding the mean and covariance mechanisms apart is needed.

A related interpretation problem appears in S1 and S3. Replacing fixed 200-marker intervals with LD-aware boundaries reduces the number of QTL-free detections. In the recorded paired checks, dense LOCO’s mean count changes from 1.33 to zero across three S1 seeds; the single paired S3 check changes from 17 to 11. This is consistent with large-QTL signal crossing arbitrary boundaries. It does not establish that every remaining detection is an error: the background is polygenic, and alternative block definitions also change the unit of discovery. Methods for constructing approximately independent LD blocks are useful here (Berisa and Pickrell, 2016; Privé, 2022), but evaluation must still define which biological or statistical null a “false locus” is meant to represent.

4.4 Correctness and computational evidence

The current review recorded 117 passing local tests. A separate installed wheel check on Python 3.10 with NumPy 1.26.4 and SciPy 1.11.4 passed 71 tests, with five skips for unavailable optional HDF5/plotting components. These checks include independent numerical references and reproductions of specific defects. The corrected contracts cover PLINK coding, sample alignment and allele counts, BLUP scaling, variance fitting, weighted kinships, window complements, permutation geometry, and effect/standard-error export. Passing them supports those contracts; it cannot substitute for an end-to-end calibration study.

One allocation change has a directly recorded before/after comparison. For 800 samples, 1,000 markers, ten LOCO groups, and 80 right-hand sides, five warm matrix-product repetitions gave identical output arrays. The measured peak allocation fell from 7,235,824 to 2,625,120 bytes, a 64% reduction. This measurement uses Python allocation tracing and excludes the pre-existing genotype cache and native BLAS allocations; it is not whole-process peak resident memory. The short timings do not support a general runtime-speedup claim.

Older timing archives establish why batching and streaming are useful, but not current competitive performance. One internal comparison recorded 0.27 seconds for a batched EMMAX scan versus 33.7 seconds for a per-variant least-squares reference at $n = 2,000$, $m = 10,000$ (about 123-fold). Historical S3 times were 221 seconds for 25-group dense LOCO and 18 seconds for the infinitesimal two-step path; a 2.7-second non-LOCO scan excluded its preceding fit. These workloads differ in both statistical procedure and accounted work. They were measured on a development laptop under concurrent activity, not a controlled cross-program benchmark.

At $n = 10,000$, a historical stochastic-Lanczos fit obtained $\delta = 0.91724$ versus 0.91639 for a full fit, a 0.09% difference in one case. That agreement concerns one nuisance estimate. A historical scan based on a truncated spectrum still produced $\lambda_{GC} = 1.30$, versus 0.85 for the full non-LOCO scan. Neither value establishes nominal calibration in that signal-containing simulation. Accurate likelihood approximation, accurate association denominators, and calibrated tests are separate requirements.

5 Discussion

5.1 The central distinction is between geometry and inference

Equation (7) explains the observed conditional gradient without invoking a software bug: covariance precision weights variant directions differently. Scalar calibration discards that distinction. The new experiment shows that restoring a few leading directions can substantially reduce the error when population structure is concentrated in those directions and the remaining spectrum is comparatively benign. It also shows why a good average rejection fraction is insufficient.

The result has a deliberately narrow scope. The leading subspace is known exactly, the covariance is fixed, genotypes contain no local LD, and the tested statistic is linear in a Gaussian phenotype before squaring. Estimated covariance, approximate eigenvectors, rare-allele sampling, nonlinear shrinkage, and phenotype-dependent selection all introduce questions not answered by the conditional calculation. The new results justify further evaluation of the spectral correction; they do not justify treating it as a generally calibrated replacement for established GWAS procedures.

5.2 Prediction accuracy and test validity need separate targets

A flexible polygenic predictor can reduce residual variance and thereby improve association power. Yet changing the predictor also changes the null distribution of a test based on its residuals. Calibration inherited from an infinitesimal model may not survive a nonlinear prior, tuning selection, or incomplete optimization. LD Score regression provides a framework for separating components of genome-wide inflation under its assumptions (Bulik-Sullivan et al., 2015); an intercept-based rescaling is not a per-variant guarantee in an arbitrary structured panel.

The same separation applies to reported prediction performance. An internally tuned or in-sample BLUP is not a prospective prediction validation. Any future prediction claim needs a held-out sample design that respects family relationships, ancestry, and the intended target population. A model can improve prediction while leaving a poorly calibrated association statistic, and a calibrated test can sacrifice power through an unsuitable background model.

5.3 The historical evidence changes the questions, not the standard of proof

The environmental experiment illustrates the noncentral mean term in Equation (10). An exposure generated from deme membership is causally external to the candidate variants in the simulation, but it is not statistically independent of them. Calling it independent would remove the very confounding mechanism being studied. A PC defined from the tested genotypes is also a legitimate mathematical stress axis, but answers a different question from a separately generated exposure. Future studies should include both with clearly stated causal and statistical interpretations.

Similarly, a locus label is a scientific choice. A block without a large QTL can tag a QTL across its boundary or contain its own small generating effects. Counting all such discoveries as false positives confounds localization, power to detect diffuse signal, and test calibration. The report therefore keeps strict-null experiments separate from QTL-free-block diagnostics. Real-data significance counts cannot resolve this ambiguity without additional validation.

5.4 Limitations and intended use

The current contribution is strongest as a transparent research implementation and a framework for controlled comparisons. The new study contains three small fixed panels, while the broader archived simulations have three or six replicates and predate important fixes. No valid matched-input external comparison is retained. There is no new validation in human multi-ancestry cohorts, no demonstrated biobank-scale resource profile, and no validation for imputed dosages or strongly imbalanced binary traits. The latter require attention to the likelihood and tail approximation, as illustrated by SAIGE’s logistic mixed-model and saddlepoint approach (Zhou et al., 2018).

The package exposes useful building blocks, but its multiple modes should not be presented as interchangeable. Dense plug-in tests, iterative infinitesimal tests, and nonlinear two-step heuristics differ in what is approximated. Reporting the mode, grouping, covariance estimation, convergence

diagnostics, and genotype definition is necessary to make a result interpretable. The most productive next step is to reduce the gap between those explicit contracts and independently validated end-to-end behaviour.

6 Future research: hypotheses and decisive experiments

The following directions are proposed work, not completed validation. Their order reflects dependence: reliable input identity and reproducible reference comparisons are prerequisites for interpreting a new method’s power or speed. Each direction has a hypothesis and a result that could force it to be revised.

6.1 Priority 1: establish a matched-input reference benchmark

Question. After input handling and nuisance definitions are matched, do the local implementations reproduce an independent Gaussian reference, and which differences from external programs remain?

The first benchmark should export one validated genotype matrix and read it back independently, checking individual hard calls, missingness, sample order, allele orientation, and variant filters before any phenotype analysis. A frozen manifest should include the marker set, covariates, phenotype, LOCO map, software versions, commands, and resource limits. Dense known-covariance and fitted-covariance cases should be separate. Comparisons with reference BOLT-LMM, LDAK-KVIK, REGENIE, and Quickdraws should document differences in model and default preprocessing rather than assume they implement the same test.

Decision criterion. Numerical discrepancies under an identical Gaussian model must be explained at the effect, standard-error, variance-component, and statistic levels before aggregate concordance is accepted. A flat loading gradient in a reference program where the local approximation shows one would restrict the claim to that local procedure. More discoveries alone would not establish better calibration or power. The invalid archived export must not be reused as the matched input.

6.2 Priority 2: test calibration independently of adaptive choices

Hypothesis. An approximation selected on calibration markers retains its claimed accuracy on independent audit markers and independent genotype panels.

A nested design should choose spectral width and scaling using one set of markers and reserve another for evaluation. For LD-containing data, hold out whole blocks where feasible, because random marker splits can make an audit nearly redundant with training. If phenotype-dependent selection is retained, repeat the entire selection and fitting procedure for every null phenotype. Pre-specified strata should include frequency, missingness, LD score, ancestry loading, kinship leverage, and their most relevant intersections. Strict-null traits should first be drawn from a known covariance; fitted-covariance experiments should then add variance estimation as a separate factor.

Decision criterion. Before running the study, specify acceptable relative rejection error at each threshold and use uncertainty intervals across independent phenotype and genotype replicates. A possible initial criterion is a rejection ratio within 0.9–1.1 at $\alpha = 0.01$ in every adequately powered prespecified stratum; it is a proposed tolerance, not a standard already met here. Genome-wide tails need their own error budget and either sufficiently large independent simulations or a justified rare-event method. Bulk calibration or a 3% training-ratio spread cannot stand in for that evidence.

Persistent failures in an audited stratum should trigger a more accurate denominator or an explicit unsupported-case result.

6.3 Priority 3: replace spectral heuristics with computable error bounds

Hypothesis. A hybrid method can retain streaming efficiency while bounding denominator error for difficult variants.

Suppose the omitted eigenvalues are known to lie in $[a, b]$, with $a + \delta > 0$, and let $s_j = \|\mathbf{z}_j\|^2 - \sum_{\ell \leq k} a_{\ell j}^2$. The exact retained contribution $q_{j, \leq k}$ then yields

$$q_{j, \leq k} + \frac{s_j}{b + \delta} \leq q_j \leq q_{j, \leq k} + \frac{s_j}{a + \delta}. \quad (11)$$

This interval can be propagated through Equation (8), or through a fixed score numerator, to determine whether an approximation is adequate at the requested threshold. Variants with wide intervals could receive additional spectral work or an iterative exact-denominator solve. Approximate Ritz values alone do not certify the spectrum: residual bounds and uncertainty in the retained subspace must be included. Equation (11) states a possible mathematical contract, not an implemented certificate.

Iterative solves offer a complementary bound. For $A = \mathbf{K} + \delta \mathbf{I} \succeq \delta \mathbf{I}$, approximate solution $\widehat{\mathbf{w}}$, and residual $\mathbf{r} = \mathbf{y} - A\widehat{\mathbf{w}}$,

$$|\mathbf{z}_j^\top (A^{-1}\mathbf{y} - \widehat{\mathbf{w}})| \leq \frac{\|\mathbf{z}_j\| \|\mathbf{r}\|}{\delta}. \quad (12)$$

A corresponding solve with right-hand side $\mathbf{z}_j$ bounds denominator error. Such bounds may be loose, especially for small δ , but make the numerical target explicit.

Decision criterion. Compare bounds with dense references on small problems, then record how often fallback solves are required, whole-process peak memory, and end-to-end time. The approach is useful only if the bound is honest and selective enough to save work. Even a perfect numerical certificate would leave estimated-covariance and model-misspecification questions unresolved.

6.4 Priority 4: derive calibration for nonlinear polygenic residuals

Hypothesis. Mixture and elastic-net residuals need a calibration that reflects their own response to the phenotype, rather than a transferred infinitesimal factor.

Start with a sequence of models whose complexity increases: fixed ridge shrinkage, tuned ridge, a fixed mixture prior, and then jointly tuned nonlinear models. For each, compare a conditional analytic or influence-function approximation where available with a parametric bootstrap that repeats fitting and hyperparameter selection. Sample splitting or cross-fitting, combined with candidate-region exclusion, may reduce reuse of information between nuisance fitting and testing, but does not by itself establish a null law. Null simulation should include both correctly specified and deliberately misspecified effect distributions, with convergence tolerances varied independently of prior choice.

Decision criterion. An apparent power gain is retained only if it survives equal-calibration comparisons at matched type-I error and can be separated from incomplete optimization or reuse of the same data for tuning. Failure of the infinitesimal factor after nonlinear fitting would support method-specific calibration, not a universal multiplicative repair. Reference-program comparisons are essential because an error in a local reimplementation is not a property of the published method.

6.5 Priority 5: separate mean confounding, covariance error, and localization

Hypothesis. The benefit and cost of PC adjustment depend on which part of the data-generating process is misspecified, and apparent null-locus inflation depends on the discovery unit.

A factorial simulation should vary ancestry-associated exposure from zero through modest effects to the S2 stress level; covariance fit from known truth to fitted and misspecified kinships; and fixed covariates from none to measured exposure and prespecified PCs. Demographic scenarios should include recent relatives, admixture, unequal ancestry sample sizes, and LD. Generate some outcomes under a strict null and others under sparse, polygenic, and ancestry-dependent effects. Report effect bias, interval coverage, conditional rejection, calibrated power, and between-ancestry performance. PC count should be chosen without looking at the final calibration outcome, or its selection repeated within validation.

For signal-containing traits, distinguish causal-variant recovery, LD-tagged association, and block-level localization. A QTL-free block should be called a false discovery only under a null definition that actually excludes its background and tagging effects. Fixed boundaries, LD-aware boundaries, and explicitly independent simulated blocks should be compared on the same underlying genotypes where possible.

Decision criterion. If LOCO is calibrated without PCs at modest exposure levels, the strong S2 result should remain a stress-test finding. If PCs reduce bias but also remove genuine ancestry-dependent signal, that trade-off must be reported instead of optimizing λ_{GC} toward one. A block method is better only if it improves the stated localization objective, not merely because relabelling blocks reduces a count.

6.6 Priority 6: make scale and reproducibility part of method design

Hypothesis. Packed genotype access, bounded caches, and selective precision can improve practical scale without changing the statistical procedure.

Benchmark whole workflows across sample size, marker count, missingness, number of LOCO groups, and genotype storage backend. Record setup, variance fitting, prediction, testing, and output time separately, as well as total wall time, CPU time, and process peak resident memory. Warm-cache and cold-cache runs should be distinguished. Measure the cost of diagnostics and fallback calculations, rather than reporting their benefits against a baseline that omits them. Mixed-precision variants should be judged by errors in effect estimates, small denominators, and tail probabilities, not only normwise matrix error.

A useful extension would add a carefully specified continuous-dosage interface with imputation-quality handling and independent decoding fixtures. Binary traits require a separate modelling and calibration programme; simply passing zeros and ones to a Gaussian scan is not the proposed extension. Prospective prediction, if developed, should use held-out families or populations and report transportability separately from association power.

Decision criterion. A scalability claim should identify the validated data size, hardware, resource limits, and failure boundary. A speedup is scientifically useful only if the inferential target and accuracy are preserved. Release artifacts should reproduce a compact set of known-truth calculations and input round trips before any larger performance claim is made.

Data, software, and reproducibility

The repository is available at <https://github.com/bvilhjal/mixmogam>. This report describes package version 2.0.0.dev1; the current numerical experiment uses baseline commit 729c8532. Its

self-contained driver is `benchmarks/denominator_geometry.py`, with results, source snapshot, and manifest in `20261003-manuscript-geometry` beneath `benchmarks/results/`. The review archive is `20261003-critical-review`. Historical inputs retain their original source snapshots. The excluded external-program archive is also preserved for provenance; its results are not incorporated into the revised displays.

`report/README.md` gives the reproduction commands, exact archive identities, and evidence map. The figure generator reads recorded results and does not silently rerun old analyses. The figure and report manifests identify source hashes; the bibliography metadata records verified DOI identities. No new human-participant data were collected for this revision. The new experiment is entirely simulated. Historical RegMap results are not used to support current performance claims.

A Useful identities and interpretation checks

A.1 Whitening explains the known-shape $\mathbf{F}$ reference

Let $L = \mathbf{H}^{-1/2}$ and let Q span $L\mathbf{X}$ orthonormally. The whitened residual phenotype is $\mathbf{r} = (\mathbf{I} - QQ^\top)L\mathbf{y}$ and the residual candidate is $\mathbf{v}_j = (\mathbf{I} - QQ^\top)L\mathbf{z}_j$. Under the known-shape Gaussian null, the squared projection onto $\mathbf{v}_j$ and the remaining residual sum of squares are independent scaled chi-squares with one and $n - p - 1$ degrees of freedom. Their ratio is Equation (5). When L is estimated from $\mathbf{y}$, this fixed-linear-transformation argument no longer proves the reference law. It explains precisely where the plug-in approximation enters.

A.2 Residual coordinates for permutation

Choose an $n \times (n - p)$ orthonormal matrix B spanning the complement of $L\mathbf{X}$. With known Gaussian covariance shape,

$$\mathbf{t} = B^\top L\mathbf{y} \sim N(0, \sigma_g^2 \mathbf{I}_{n-p}), \quad \text{Var}\{(\mathbf{I} - QQ^\top)L\mathbf{y}\} = \sigma_g^2(\mathbf{I} - QQ^\top). \quad (13)$$

The entries of $\mathbf{t}$ are exchangeable; the n entries of the projected residual generally are not. A valid residual-coordinate transformation therefore permutes $\mathbf{t}$ and maps it back through B , rather than permuting arbitrary sample-coordinate residuals. The implementation uses orthogonal transformations without storing an additional full complement matrix in the production path. Estimated covariance, non-Gaussian errors, and repeated model selection require separate investigation; algebraic exchangeability under a known model should not be advertised as an unconditional guarantee.

A.3 BLUP scale and a marginal Bayes factor

The genetic conditional mean uses $\text{Cov}(\mathbf{u}, \mathbf{y}) = \sigma_g^2 \mathbf{K}$ and $\text{Var}(\mathbf{y}) = \sigma_g^2 \mathbf{H}$, hence

$$\hat{\mathbf{u}} = \sigma_g^2 \mathbf{K}(\sigma_g^2 \mathbf{H})^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{K}\mathbf{H}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}). \quad (14)$$

This identity is conditional on fitted parameters when used in the package. In-sample fitted genetic values do not measure prospective prediction accuracy.

For a marginal effect estimate $\hat{\alpha}$ with sampling variance v , assume $\hat{\alpha} \mid \alpha \sim N(\alpha, v)$ and an alternative prior $\alpha \sim N(0, W)$. Against a point null at zero, the approximate Bayes factor is

$$\text{BF}_{10} = \sqrt{\frac{v}{v + W}} \exp \left\{ \frac{\hat{\alpha}^2 W}{2v(v + W)} \right\}. \quad (15)$$

The prior variance W has the units of the squared effect and must be chosen explicitly. Posterior odds additionally require prior odds; normalizing marginal factors across correlated variants does not by itself yield a multi-variant fine-mapping posterior. The corrected API therefore no longer presents an RSS-based pseudo-score as a posterior probability.

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